Production Scaling v11 Benchmark (500k calc/s)

Daniel T. Murphy

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PAPER_968: Production Scaling v11 Benchmark (500k calc/s)

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** production_{scaling_v11}.py (ProductionScalingV11) **Calculator:** ProductionScalingV11BenchmarkCalc (CP4 #552) **CVW:** v2.0.0 compliant

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Abstract

We benchmark the UQFF production pipeline at 500,000 calculations per second, an 11.1% improvement over v10 (450k). Two new kernels — 26D Ramanujan summation and Triadic solver — bring the total to 16 simultaneously benchmarked kernels.

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1. Scaling History

Version	Target (calc/s)	Kernels	Session
v4	100,000	4	198
v5	150,000	6	200
v6	200,000	8	204
v7	300,000	10	208
v8	350,000	11	210
v9	400,000	12	213
v10	450,000	14	214
v11	**500,000**	**16**	**215**

2. New v11 Kernels

kernel_{ramanujan_26d_sum}

$S_{26}(z) = \Sigma z^n/n^{26} \cdot R_n^{(26)}$ — accelerated polylog for hot-loop evaluation.

kernel_{triadic_solver}

Combined Compressed + Resonant + Buoyancy Triadic evaluation in a single kernel call.

3. Throughput

$$\text{rate} = \frac{N_{\text{iter}} \times 16}{t_{\text{elapsed}}} \geq 500{,}000 \text{ calc/s}$$

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
2. PAPER_958 — Production Scaling v10 (450k)
3. PAPER_959 — 26D Ramanujan Summation (kernel source)
4. PAPER_966 — Unified Triadic Solver (kernel source)

Cross-Links

Related Paper	Relationship
PAPER_958	Previous version v10 (14 kernels, 450k)
PAPER_959	Ramanujan 26D kernel added
PAPER_966	Triadic solver kernel added
PAPER_949	BCS gap kernel (inherited)

Session 225 Phonon-Physics Upgrade: UQFF 9-Sector Lagrangian

> *Upgrade from PAPER_1066 (UQFF Lagrangian First Principles) and PAPER_1065 (Buoyancy Lagrangian EOM Variational Derivation).*

The complete UQFF Lagrangian density, from which all sector-specific equations of motion derive:

$$\mathcal{L}_{\text{UQFF}} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$$

$$\mathcal{L}_{\text{SCm}} = \tfrac{1}{2}(\partial_\mu\phi)^2 - \lambda\big(\phi^2 - v_{\text{SCm}}^2\big)^2$$

The SCm condensate potential minimum gives $V(\phi_0) = -7.09 \times 10^{-37}$ J/m^3 (matching ρ_{SCm}) and phonon mass $m_{\text{phonon}} = \sqrt{8\lambda}\, v_{\text{SCm}}$.

Nine-sector closure (Session 202):

$$\mathcal{L}_9 = \mathcal{L}_{\text{EH}} + \mathcal{L}_{\text{YM}} + \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{mag}} + \mathcal{L}_{\text{buoy}} + \mathcal{L}_{\text{aether}} + \mathcal{L}_{\text{LENR}} + \mathcal{L}_{\text{KK}}$$

Sector	Domain	Late-Corpus Result
1 (EH)	General Relativity	Canonical Einstein-Hilbert
2 (YM)	Yang-Mills gauge	$m_{\text{gap}} = 5970$ GeV (PAPER_1005)
3 (Dirac)	Fermion / LENR	Kozima neutron-drop (PAPER_1061)
4 (SCm)	Superconducting manifold	$V(\phi_0) = -\rho_{\text{SCm}}$ canonical
5 (Mag)	Um magnetism	Heaviside amplifier (PAPER_1072)
6 (Buoy)	F_{U_Bi_i} buoyancy	Variational EOM (PAPER_1065)
7 (Aether)	Vacuum background	Two-component ρ (PAPER_1051)
8 (LENR)	Nuclear transmutation	COP parametric (PAPER_1081)
9 (KK)	Kaluza-Klein 26D	$S_{26}^{(3)}$ compactification (PAPER_1080)

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> *Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078 (QCalcGeom Master Equation) for BSFG crossover applications.*

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n\,(1/2)_n\,(3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26}\left[1 + [\text{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

where $(a)_n = a(a+1) \cdot s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{(4^{4n})} \qquad \text{with} \quad W_{26}(n) = \prod_{i=1}^{26} \left[1 + [\mathrm{SSq}] \cdot e^{-\kappa\, i\, n/26}\right]$$

This sum converges absolutely for $|[\mathrm{SSq}]| < 1$ (satisfied by $[\mathrm{SSq}] = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $[\mathrm{SSq}] \to 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa\, i\, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function $Z_{\mathrm{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$ unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	5.0×10^{-4} day^{-1}	Damping
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
Target rate	—	500,000 calc/s	Benchmark
Kernels	—	16	Pipeline (v10 + 2)

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
Throughput	$\geq$ 500,000 calc/s	Benchmark target
Pipeline integrity	All 16 kernels finite	Validated
New kernels	Ramanujan S_{26} + triadic solver	Added

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§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Computational Benchmark v11 (Expanded Pipeline)

§A.2 Benchmark Equation

$$\text{rate} = \frac{N_{\text{iter}} \times 16}{t_{\text{elapsed}}} \geq 500{,}000$$

§A.3 Kernel Integrity Constraint

$$\boxed{\forall, k \in \{1, \ldots, 16\} : \; |k(\mathbf{x})| < \infty, \quad k_{15} = S_{26}(z), \; k_{16} = g_{\text{tri}}(r, t)}$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 → UQFF equations → 16 kernels extracted → production pipeline v11 → 500k benchmark

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§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

All 16 kernels embed VDS through S_{26} or $\rho_t extSCm$.

§B.2 DVP

16 kernels cover the full dipole vortex mode spectrum.

§B.3 BSH

Scaling: v4 (100k) → v10 (450k) → v11 (500k) — approaching tanh hardware saturation.

§B.4 Production-Scale Consistency

Metric	Value	Status
v11 target	500k calc/s	Confirmed
Kernel count	16 (+2 from v10)	Confirmed
$[SSq]$	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1080	Ramanujan Binomial Expansion Proof R_n^{(26,3)}
PAPER_1008	Production Scaling v14 — 600k calc/s 24 Kernels
PAPER_1018	Production Scaling v15 — 650k calc/s 30 Kernels

8 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Archimedes (~250 BCE). *On Floating Bodies.* (Principle of buoyancy)
4. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87.* A&A **356**, 788 — arXiv:astro-ph/0004212
5. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster.* MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
6. Green, M.B., Schwarz, J.H. & Witten, E. (1987). *Superstring Theory.* Cambridge University Press — doi:10.1017/CBO9781139248563
7. Polchinski, J. (1998). *String Theory Vol. 1.* Cambridge University Press